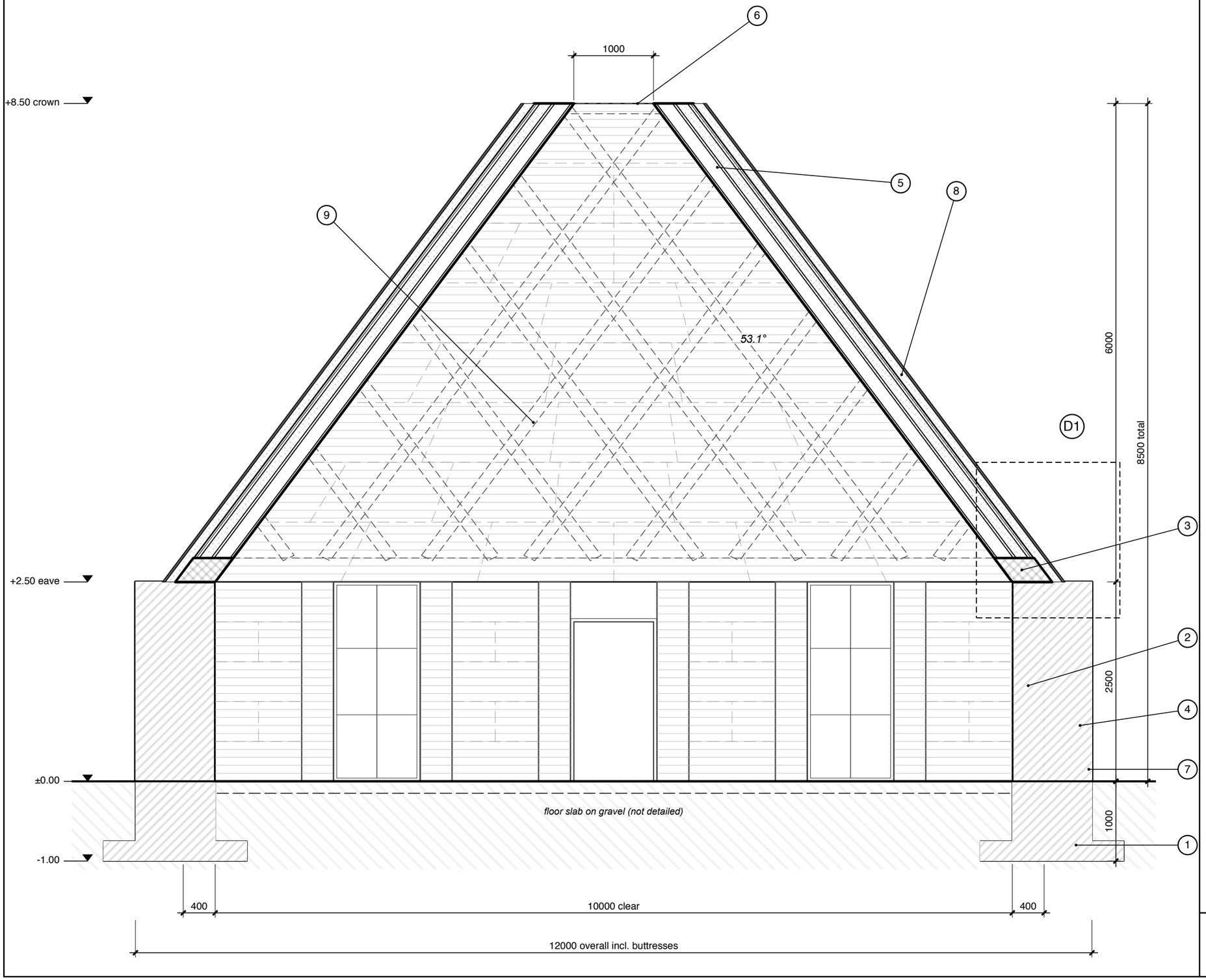
Total section A-A · grid B

through the east and west buttresses, looking north



1. Footing 1800 x 250, underside at -1000 (frost depth and bearing are guesses).
2. Wall 400. The cut passes through a buttress, so the wall is solid with the buttress here (no insulation). Between buttresses it is printed: an inner bead 30, a lattice zone 310 (wavy bead + insulation), 2 outer beads 30.
3. Solid base of the roof: the first 300 (vertical) above the eave is completely solid concrete, 400 wide, no cavity or lattice zone; there is no ring beam.
4. Buttress 400 wide, 1000 from the inner face (600 beyond the wall), one about every 1.5 m (8 per side); it rises to +2.50, flush with the roof base.
5. Roof shell 400 thick square to the slope, the same from eave to crown (about 500 horizontally), every printed layer one 30 bead: inner bead 30, rib zone 160 (cavity crossed by the cast ribs), middle bead 30, lattice zone 120 (wavy bead + insulation, dotted), 2 outer beads 30.
6. Crown opening 1000 clear; roof rises 6000 over 4500, slope 53.1° (36.9° from the vertical).
7. Everything is 3D printed in 10 mm layers; crossed hatch = solid cast concrete (roof base, ribs). Printed layer joints and the cast fill of the buttress and footing are shown in the roof-grid sheets and in detail D1.
8. Outside the printed shell: insulation 100 (dotted), then the outer layer (cladding 30, dark).
9. Cast ribs of the roof grid, seen on the back face as dashed outlines (roof grid B, diamond lattice). The cut lies between ribs, where the rib zone is an empty cavity; rib sections are on the roof-grid sheets.

Notes

Roof build-up as in the Karamba model (scripts/karamba_model.py): bead 30, 2 outer beads, lattice 120, rib zone 160; inner and middle bead are the structural skins, the ribs the webs between them.

The cut runs through the 4th buttress from the south (y = -743), as on the floor plan. Beyond it the north wall is seen as on the plan: buttress every 1.5 m (strips outside the wall), every other bay printed wall, the bays in between full-height curtain wall (door in the middle one).

Faint grid on the back roof face and the north wall = printed pieces in running bond, about 1500 wide (assumed sizes). Dashed outlines on the back roof face = cast ribs.

0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

Scale 1:50 at A3 · dimensions in mm

2026-10-14 · hand estimates, not engineered

05